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A Theoretical Framework for Supporting Clustering Validation via Non-Negative-Matrix-Factorization Trace Sequences over Probabilistic Spaces

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***Abstract*—This paper focuses the attention on the annoying problem of *clustering validation*, intended as a relevant task of major analytical processing. The proposed method argues to build and analyze a *data matrix from probabilistic spaces* and to exploit *Non-negative Matrix Factorization* (NMF) to identify existing clusters and select them according to a *Bayesian interpretation*.**

***Index Terms*—Clustering Quality, Clustering Validation, Probabilistic Clustering Validation, Advanced Clustering Analysis**

I. Introduction

Clustering algorithms are utilized in various topics and their applications are growing at an immediate pace [20], since the development of classic k-means [28]. Clustering basically follows the goal of discriminating similar data to put them in the same group which is different from the other groups according to their characteristics. It categorizes clustering methods into two main scenarios: (*a*) *hard clustering*, in which groups do not have any overlap, and (*b*) *soft clustering* which groups may have overlap with each other.

In clustering techniques, finding a suitable number of clusters is an important issue. This issue is called *cluster validation*, and there are many approaches invented up to now to address it [1], e.g. some of them use sensitivity to solve this problem [1], [16]. This problem is becoming very important, for instance in *big data analytics* (e.g., [8], [29]).

The process of validation can involve creating a *loss function*, labeled as $\mathcal{F}$, which has the goal of minimizing the distance between observations and statistics that represent the number of groups (clusters). This optimization procedure yields parameters for the most accurate estimate of the clustering solution. Following that, two effective validation approaches are introduced: the *Silhouette index* and the *Gap statistics*. The Silhouette index gauges the similarity of an object to its own cluster in comparison to other clusters, thereby providing a measure of separation between clusters [34]. A higher Silhouette value indicates well-defined clusters. In contrast, the Gap statistics compare the scattering within clusters to the expected scattering of a reference distribution, which helps in evaluating the quality and significance of clustering [38]. These techniques offer valuable insights for assessing clustering algorithms and determining the optimal number of clusters for a given dataset.

In [18], the authors introduce an approach based on soft clustering that are utilizing an indicator function for observations by employing *kernelization*, where the null hypothesis acts as a classifier. Another technique [36] applies likelihood cross-validation to estimate the number of model components. The approach introduced by [33] utilizes the χ^2 statistics to assess cluster similarity. Additionally, [40] applies an algebraic method to cluster the data points based on collinearity. This area obtained a special focus after the emergence of the coronavirus pandemic [27], [35] likely exploring clustering in pandemic-related data. Furthermore, a review paper [39] centers on bioinformatics and its association with soft clustering techniques. Although the intricacies of each method remain somewhat mysterious, these references offer a glimpse into the diverse range of soft clustering techniques and their applications in various domains.

In this paper, we introduce a technique for validating the optimal number of clusters. The method involves analyzing the *data matrix from a probabilistic space* and utilizing *Non-negative Matrix Factorization* (NMF) to identify existing clusters. By generating a sequence of traces for different dimensions of the non-negative approximation and observing its limit which is a *Gamma distribution*, the approach provides valuable insights into the credibility and probabilities of selecting the suitable number of clusters, in a *Bayesian interpretation*. This paper completes and extends the previous short paper [15], where we introduced the main clustering validation technique.

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II. Related Work

In the literature, the utilization of generative models within the framework of clustered data sets particularly occurs in scenarios where *probabilistic classification* is adopted. Two fundamental notions of probabilistic clustering are delineated: (*a*) The initial notion postulates varying probabilities for the affiliation of individual entities to distinct clusters, as expounded upon in [13]. (*b*) The secondary notion revolves around the classification of observations into clusters to which they exhibit the highest likelihood of belonging, as elucidated by [11]. The following equation represents the suitable mixtures or *components* which both methods follow in number and parameter:

$$f(a|\kappa) = \sum w_j d_j(a|\kappa_j) \tag{1}$$

where a is a data vector of the recorded input sample data matrix, d is a density with parameter κ, and the combined weights associated with each density are designated as w in which the sum is operated on all of the combinations. This approach converts an unsupervised classification problem into *a parameter estimation challenge*. This involves figuring out the probability that each item within a cluster actually belongs to that cluster. The accuracy of the classification needed to evaluate the likelihood of different cluster compositions is referred to as *credibility* or *likelihood*. In other words, this method computes the posterior probability of the number of combinations to evaluate the quality of the classification [36].

Several studies, such as [18], present an indicator function for observations based on kernelization, and employ the null hypothesis as a classifier, which is responsible for a number of research that have concentrated on this perspective of probabilistic clustering for validation challenges. To determine the number of model components, likelihood cross-validation is utilized in [36]. By employing an alternative method, the χ^2 statistics comparing probabilistic classifications ([33]) may be used to assess the similarity between clusters.

The introduction of an index and the assumption of a parametric distribution for each cluster, as outlined in the prior investigation [17], can yield a validation index. Recent studies, including [32], center their attention on astronomical observations and make the assumption of normality and the presence of correlations. In their study, they present an algorithm in which the posterior distribution of correlations adheres to a Gamma probability distribution. An alternative strategy put forth by [40] proposes an algebraic technique to cluster data points based on their *collinearity*.

A review paper [11] mentions that methods based on NMF can be utilized in probabilistic classification problems due to the probabilistic essence of the NMF. The introduction of NMF techniques for clustering is attributed to the work [12]. These techniques involve probabilistic classifications and can be viewed as analogous to k-means with the incorporation of a *Bayesian classifier*. Despite this equivalence, the unique characteristics resulting from these techniques have not been utilized for validation purposes in the same manner as in the present study.

In [37], the authors present *Ensemble Cluster Validity Index* (ECVI), a novel approach that aims to determine the optimal number of clusters by integrating and optimizing multiple clustering validity indices (e.g., Silhouette index, *Davies–Bouldin* index, *Calinski-Harabasz* index, the Gap statistic, etc.). The primary objective of the ECVI is to improve the assessment of dataset partitioning correctness, ensuring it accurately represents the underlying structure of the data. In order to achieve this, the clustering solution obtained through the ECVI is utilized as a crucial input parameter for the *k-means clustering* algorithm. Essentially, the ECVI focuses on the development and validation of an internal validity method to identify the most appropriate NC for the given dataset. The experimental evaluation demonstrates that the proposed ECVI consistently outperforms all compared indices in terms of both optimal NC selection and accuracy rate.

[30] introduces a novel guiding principle, *a high-quality clustering should exhibit stability, but it should also exhibit the absence of stable partitions within each cluster*. This principle serves as the foundation for a novel clustering validation criterion, which leverages both *between-cluster* and *within-cluster* stability metrics. This innovative approach overcomes the constraints associated with previous stability-based methods. In order to prove the effectiveness of the novel criterion in selecting the appropriate number of clusters, they elaborate a comprehensive comparative analysis, pitting the proposed criterion against existing methods to highlight its advantages and superior performance in the realm of clustering validation.

[31] focuses on *addressing missing values in both clustering and cluster validation* processes by developing a novel toolbox designed to tackle this challenge comprehensively. This toolbox encompasses a diverse array of algorithms tailored for data pre-processing, distance estimation, clustering, and cluster validation, all with the explicit consideration of missing values. In this work, they delve into the methodological underpinnings of the algorithms integrated into the proposed toolbox, elucidating their theoretical foundations and practical applications that showcase its efficiency to generate robust and meaningful clustering results in the presence of missing values. Evaluation findings confirm the benefits of the proposal.

III. Building and Manipulating Data Matrices from Probabilistic Spaces

We define a data matrix as $\mathbf{A} \in \mathbb{R}_+^{q\times v}$, where q is the number of samples indexed by i, and v is the number of *iid* variables which each variable is indexed by j. The values of the data are positive real values but there isn't any statistical hypothesis. With this definition, a row vector of the data matrix is considered as $\mathbf{a}_i = (a_{i1}, \ldots, a_{iv})$ that refers to each record, and the column vectors are considered as $\mathbf{a}_j = (a_{j1}, \ldots, a_{jq})'$ (Note that the columns are linearly independent), where their elements are the values of $\mathbf{a}$ on the variable j.

We define a transformation of $\mathbf{a}$ as follows:

$$[\mathbf{B}]_{ij} = [\mathbf{A}]_{ij}\,\mathbf{C}_A\,\mathbf{N}_A \qquad (i = 1, \ldots, q \;\text{ and }\; j = 1, \ldots, v) \tag{2}$$

where $\mathbf{C}_A$ and $\mathbf{N}_A$ are diagonal matrices of suitable dimensions considered as:

$$\mathbf{N}_A = \operatorname{diag}\Big(\sum_j a_{ij}\Big)^{-1} \tag{3}$$

$$\mathbf{C}_A = \operatorname{diag}\big(q\big)^{-1} \tag{4}$$

are constant when measurement scale changes, and $\sum_{ij}\mathbf{B} = 1$.

To obtain the matrices $\mathbf{U}$ and $\mathbf{G}$, we need to pose the NMF on matrix $\mathbf{B}$ in a way that the Equation (5) minimizes on some norm or divergence [5], with the condition $\sum_{ij}\mathbf{VG} = 1$.

$$\begin{aligned}[\widehat{\mathbf{B}}]_{ij} &= [\mathbf{U}]_{ik}[\mathbf{G}]_{kj} \qquad\qquad (k = 1, \ldots, k) && (5)\\ &\approx [\mathbf{B}]_{ij} && (6)\end{aligned}$$

Here we consider the *Kullback-Leibler (KL)* divergence [22].

$$C_{KL}([\mathbf{B}]_{ij} \| [\mathbf{UG}]_{ij}) = \sum_{ij} [\mathbf{B}]_{ij} \circledast \log\frac{[\mathbf{B}]_{ij}}{[\mathbf{U}\,\mathbf{G}]_{ij}} \tag{7}$$

Karush-Kuhn-Tucker (KKT) conditions are exploited to optimize the above-mentioned equation, and that leads to the solution presented in the following [14]:

$$[\mathbf{U}]_{ik} \leftarrow [\mathbf{U}]_{ik} \circledast \left(\frac{[\mathbf{B}]_{ij}}{[\mathbf{U}\,\mathbf{G}]_{ij}}[\mathbf{G}]'_{kj}\right) \tag{8}$$

$$[\mathbf{G}]_{kj} \leftarrow [\mathbf{G}]_{kj} \circledast \left([\mathbf{U}]'_{ik}\frac{[\mathbf{B}]_{ij}}{[\mathbf{U}\,\mathbf{G}]_{ij}}\right) \tag{9}$$

where the product and fraction are both element-wise, called *Hadamard*, which means the operations apply on equal indices of matrices' elements and the $\circledast$ refers to the element-wise product.

In this step, firstly we choose a value for k, then the goal is to find an appropriate answer for $\mathbf{U}$ and $\mathbf{G}$ using an iterative approach on the Equation (8) and Equation (9) until the following condition is satisfied:

$$\Big\|[\widehat{\mathbf{B}}]_{ij} - [\mathbf{B}]_{ij}\Big\|_1 \leq \epsilon \qquad (\epsilon > 0) \tag{10}$$

Note that, in the above-mentioned, the *Hilbert norm* L_p $(p = 1)$ is designated as $\|\cdot\|_1$, which refers to the sum of all of the records, and is different from the *Hilbert-Schmidt norm*, which is L_2 or the *Gaussian norm*. It is induced in a natural way from the sums of Equation (7).

We utilized *em* algorithm (which is mostly the same as the constantly converging approach *Expectation Maximization (EM)* algorithm) to optimize the objective function based on KL divergence [10]. Generally, EM results are asymptotes to the *em* algorithm [2].

According to *probabilistic normalization* [12] we impose appropriate normalization constraints on NMF. Considering Equation (5) the constraint expressed as $\|\cdot\|_1 = 1$ for columns of $\mathbf{B}$ and $\mathbf{U}$ and rows of $\mathbf{G}$.

In the study [12] it is shown that the probabilistic normalization results in the following equation :

$$[\mathbf{UG}]_{ij} = \frac{1}{k}[\widetilde{\mathbf{U}}]_{ik}[\widetilde{\mathbf{G}}]_{kj} \tag{11}$$

In Equation (11), we introduced the normalized matrices by tilde. Therefore, we obtain the matrices with a constant k' as introduced in the following:

$$[\widetilde{\mathbf{U}}]_{ik'} \sim P(u_i|\,k') \tag{12}$$

$$[\widetilde{\mathbf{G}}]_{k'j} \sim P(g_j|\,k') \tag{13}$$

Here, k is also represented as the *model components*. In [1], they are presented to be applied in clustering problems, in addition to exploiting the information retrieval problems with the latent variables [12].

IV. NMF Factorization Trace Sequences for Clustering Validation

A trace is defined as follows:

$$\operatorname{tr}\big(\widehat{\mathbf{B}}'\widehat{\mathbf{B}}\big) = \sum_k \operatorname{diag}\big([\mathbf{U}]_{ik}[\mathbf{G}]_{kj}\big)'\big([\mathbf{U}]_{ik}[\mathbf{G}]_{kj}\big) \tag{14}$$

In addition, it results in a sequence, designated as s, that is indexed by k in the NMF factorization space span and showed in the Equation (15) and Equation (16):

$$\begin{aligned}\widetilde{s}_k &= \Big\{\operatorname{tr}\,\big([\mathbf{U}]_{i1}[\mathbf{G}]_{1j}\big)'\big([\mathbf{U}]_{i1}[\mathbf{G}]_{1j}\big),\\ &\quad \operatorname{tr}\,\big([\mathbf{U}]_{i2}[\mathbf{G}]_{2j}\big)'\big([\mathbf{U}]_{i2}[\mathbf{G}]_{2j}\big), \ldots\Big\} && (15)\\ &= \Big\{\widetilde{s}_1, \widetilde{s}_2, \ldots\Big\} && (16)\end{aligned}$$

$$s = \operatorname{tr}\,\big([\mathbf{B}]'_{ij}[\mathbf{B}]_{ij}\big) \tag{17}$$

The following equation is derived from the quotient of Equation (16) and Equation (17):

$$s_k = \Big\{\frac{\tilde{s}_k}{s}\Big\}_k \tag{18}$$

When we apply the constraint presented in Equation (10) to all products of $\mathbf{UG}$ regarding changing k we obtain the following (only for $k' > k$):

$$\big\|[\mathbf{U}]_{ik}[\mathbf{G}]_{kj}\big\|_1 = \frac{1}{k}\big\|[\widetilde{\mathbf{U}}]_{ik}[\widetilde{\mathbf{G}}]_{kj}\big\|_1 \tag{19}$$

$$\frac{1}{k^2}\text{tr}\left([\widetilde{\mathbf{U}}]_{ik}[\widetilde{\mathbf{G}}]_{kj}\right)'\left([\widetilde{\mathbf{W}}]_{ik}[\widetilde{\mathbf{G}}]_{kj}\right) > \\ \frac{1}{(k')^2}\text{tr}\left([\widetilde{\mathbf{U}}]_{ik'}[\widetilde{\mathbf{G}}]_{k'j}\right)'\left([\widetilde{\mathbf{U}}]_{ik'}[\widetilde{\mathbf{G}}]_{k'j}\right) \tag{20}$$

A. Definition of Limit for Trace Sequences

Considering Equation (18), we define the function in Equation (21) with a descending k as a one-to-one map of the KL divergence:

$$\varphi(s_k) = \left(\frac{s_k}{s}\right)^{-s} \tag{21}$$

Now we transform the variables as Equation (22) which results in Equation (23):

$$\frac{1}{\mu} = \frac{s_k - s}{s} \tag{22}$$

$$\psi(\mu) = \frac{s}{\mu^2}\left(1+\frac{1}{\mu}\right)^{-s} \qquad \left(\left|\frac{\partial\, s_k}{\partial\, \mu}\right| = \frac{s}{\mu^2}\right) \tag{23}$$

where μ can change in the interval $s/(s_1 - s) \leq \mu < +\infty$, since $s_k \rightarrow s$ regarding increasing k, as a consequence of Equation (10) for $k > \min(q, v)$.

From Equation (23) we can obtain Equation (24) and Equation (25):

$$\psi(\mu) = \frac{s}{1-s}\frac{\partial}{\partial\mu}\left(1+\frac{1}{\mu}\right)^{1-s} \tag{24}$$

$$\left(1+\frac{1}{\mu}\right)^{1-s} \xrightarrow[\mu\rightarrow\infty]{} \exp\left(\frac{1-s}{\mu}\right) \tag{25}$$

The Equation (26) derives from replacing Equation (26) into the Equation (24):

$$\psi(\mu) = c\frac{\partial}{\partial\mu}\exp\left(\frac{1-s}{\mu}\right) \qquad \left(c = \frac{s}{1-s}\right) \tag{26}$$

After changing $a = 1/\mu$, $\eta = s - 1$, and considering the values of η to $0 < \eta < 1$, we obtain $\psi(x, \eta)$ as follows:

$$\psi(x,\eta) = c\frac{1}{a^2}\frac{\partial}{\partial\, a}e^{-\eta\, a} \qquad \left(\left|\frac{\partial\, \mu}{\partial\, a}\right| = \frac{1}{a^2} \text{ and } \left|\frac{\partial\, s}{\partial\eta}\right| = 1\right) \tag{27}$$

$$= c\,\frac{\partial}{\partial\, a}\frac{\partial^2}{\partial\eta^2}\, e^{-\eta\, a} \tag{28}$$

$$\frac{\partial^h}{\partial\, a^h}\frac{\partial^{p-2}}{\partial\eta^{p-2}}\psi(\eta, a) \; = (-1)^{h+p}c\frac{\partial^h}{\partial\, a^h}\frac{\partial^p}{\partial\eta^p}\, e^{-\eta\, x} \\ (p > 2 \text{ and } h \geq 1) \tag{29}$$

B. Normalization of Limit for Trace Sequences

Now, we rewrite the derivatives in Equation (29) to obtain Equation (30):

$$f(a,\eta) = (-1)^{h+p}c\eta^h a^p e^{-\eta\, a} \tag{30}$$

After applying normalization on Equation (31), we obtain the following:

$$f(a\eta) = \frac{f(a,\gamma,\eta)}{\displaystyle\int_0^{+\infty}\int_0^1 f(a,\gamma,\eta)d\eta\, da} \tag{31}$$

$$= \frac{1}{\Gamma(p+1)}\eta^h a^p e^{-\eta\, a} \tag{32}$$

Then, for $\gamma = h = p + 1$, we obtain the following which is Gamma PDF of parameters γ and η:

$$f(a,\gamma,\eta) = \frac{1}{\Gamma(\gamma)}\eta^\gamma x^{\gamma-1}e^{-\eta\, a} \tag{33}$$

With a Gamma PDF, the number of clusters can be described using Equation (33). This can be correct for the values $a \in \mathbb{S}_+$ that can be obtained by a base in Equation (21), but here we try to exploit a variety of criteria in statistics to remove it.

C. Loss Function Equivalence

We need to define the dimensionality of the decomposition of the space span with clusters, thus, the clustering method should use the NMF. For this goal, we estimate k densities instead of the clustering problem [11]. Therefore, the probabilities of the number of clusters are the integer values of the Gamma support PDF, additionally, if x includes positive integers the Equation (33) models a cluster density for them.

The Equation (21) is equivalent to a loss function in cases where we consider the expansion to the divergence KL in the following way:

$$C_{KL}(s_k\|\, s) = s\log\, s - s\log\, s_k \tag{34}$$

$$= G(s) - G(s_k|\, s) \tag{35}$$

where *entropies* are designated as $G(\cdot)$ and $C_{KL}(s_k\|\, s)$ can be interpreted as the *mutual information*.

For the sake of simplicity, we will not go through the *Information Theory* but only mention that the Equation (33) is a one-to-one map of a loss function of the Equation (34), or in Information Theory format as in Equation (35). Now we replace the common parameters optimization approach with an approach to achieve the limit distribution and normalize it in order to make it consistent with probability postulates.

Algorithm 1 `Traces Sequence Limit Function`

Procedure 1

Input: Data Matrix: $\mathbf{A}$

Parameters: k (components), r (re-estimations), ϵ (approximation)

Output: $\mathbf{S}$ (traces), rk (rank), s (trace of $\mathbf{B}'\mathbf{B}$)

$\mathbf{B} \leftarrow \mathbf{A}$

$rk \leftarrow \mathbf{B}$

$s \leftarrow \mathbf{B}$

for $r \leftarrow 1$ *to re-estimations* **do**

  for $k \leftarrow 1$ *to components* **do**

    Initialize $\mathbf{U}$ $\mathbf{G}$

    while $\epsilon' \geq \epsilon$ **do**

      compute $\mathbf{U}$ and $\mathbf{G}$

      $\epsilon' = \|\mathbf{UG} - \mathbf{B}\|$

      $s_k = \text{tr}(\mathbf{UG})'\mathbf{UG})$

    end while

  end for

end for

Procedure 2

Input: $\mathbf{S}$, rk, s

Output: Z_{mean} (traces), L (Basis transformation)

$L = \frac{s}{1-s}$

$L_z = L\sqrt{qv}$

support $= L_s\,(1, 1, \dots)$

for $k \leftarrow 1$ **to** components **do**

  $Z_{mean} = \sum_i \mathbf{S}_{ik}/q$

end for

Procedure 3

Input: Z_{means}

Output: γ,η

$\gamma - 1 = \max Z_{mean}$

$\eta = 1 - 1/\gamma$

V. Case Studies and Preliminary Experiments

The Algorithm 1 demonstrates three procedures that present the process of obtaining the PDF of Equation (33) with all the parameters that are needed for the process. During the first procedure, the number of components (k), defining a sequence of k terms, and the approximation degree of Equation (10) (known as ϵ), are imposed on the data for the reason that it is required to derive the sequence of traces. We produce matrices in Equation (8) and Equation (9) with random starting conditions using 3 iterations repeated 25 times to simplify the iterative procedure, rather than the approximation criteria. Additionally, a matrix called $\mathbf{S}$ includes the sequences of traces, and the rank of the matrix $\mathbf{B}'\mathbf{B}$ and its traces should be obtained.

A value for estimation of the expected value for each k is obtained, once the parameters including sequence of traces matrix $\mathbf{S}$, the rank of matrix $\mathbf{B}$, and the trace of the matrix s=$\mathbf{B}'\mathbf{B}$ are determined. In other words, the first procedure's result goes to the second procedure as its input. We pursue the target of computing the estimation of the traces on the support $\mathbb{S}_+$, which is as follows:

$$c_h = \frac{s}{1-s}\sqrt{qv_h} \tag{36}$$

here the number of linear independence variables is designated as v_h. In Figure 2 the dot lines are computed by multiplying each term of the sequence by Equation (36).

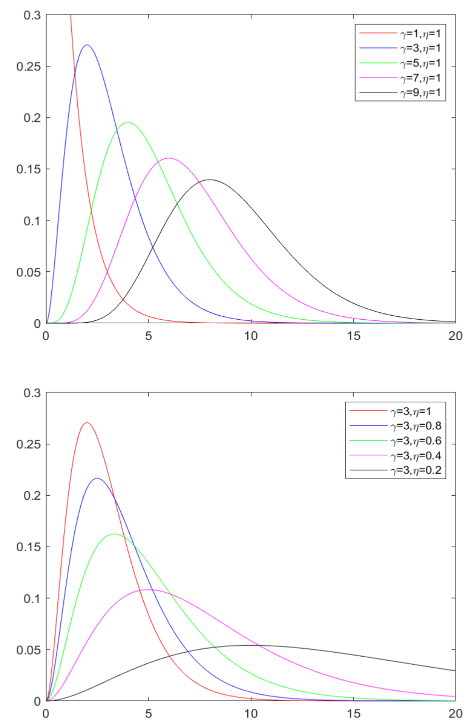

Fig. 1. Gamma PDF plots. Upper Plot Shows the Effect of Parameter γ with Fixed η. Lower Plot Fixes γ and Varies η.

In Procedure III, the main target is to estimate the parameters of Gamma PDF. It is necessary to adjust the maximum of the parameters of Gamma PDF to the supremum of Equation (21). Considering the maximum of Equation (33) equal to $(\gamma-1)/\eta$, by transforming of variable $t = \eta\, a$, the expectation of a standard Gamma distribution is γ [3]. Thus, we can write the following equation:

$$\eta = 1 - \frac{1}{\gamma} \tag{37}$$

To be more clear, Figure 1 shows the changes according to parameters γ and η. To test our method, we applied it to several data sets selected from the UCI repository[1]. The outcome is illustrated in Figure 2.

We compare the results obtained from our approach with the Silhouette index in order to demonstrate how our validation

[1] $https://archive.ics.uci.edu/ml/datasets.php$

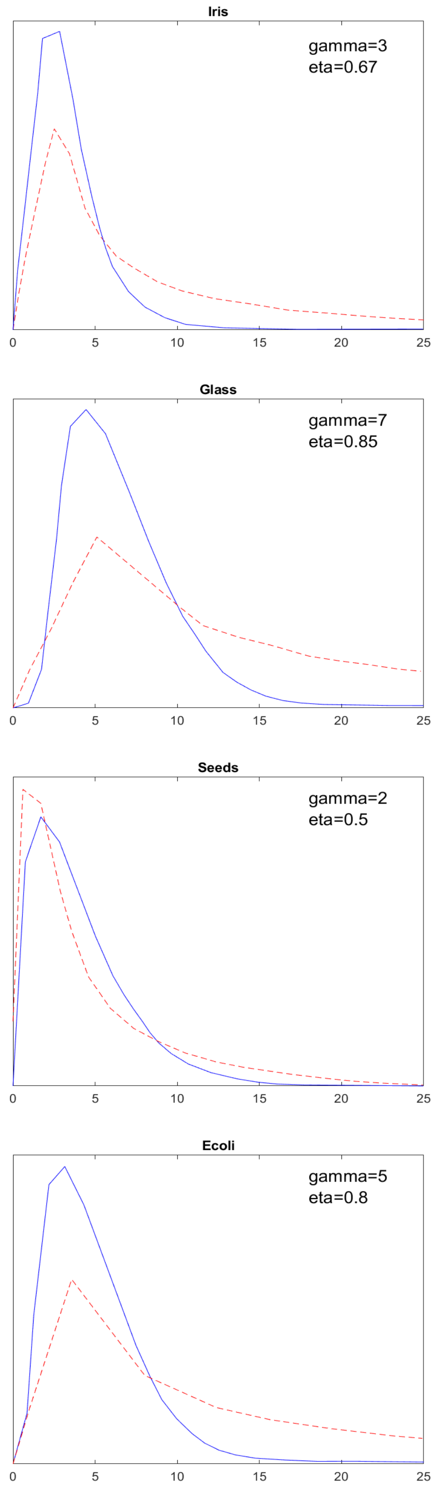

Fig. 2. Representation of the Studied Data Sets. Dashed Lines Corresponds to Equation (21) and Continuous Line are the Corresponding Gamma PDF.

criteria work. It is done by exploiting a bootstrap parameter $\beta = 70$ into the *factoextra* *R*-package [21], and the results are illustrated in Figure 3, where the values of the Silhouette

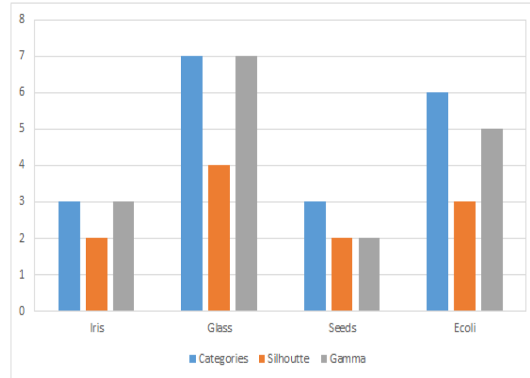

Fig. 3. Number of *Categories* Defined as the Optimal Cluster Number by the Data Set Owner; *Silhouette* and *Gamma* Values Provided by Our Approach.

index are collected as well as the PDF's maximum values (see Figure 1). The comparison shows that the values obtained from our validation criteria are superior to what was obtained from the classical index.

We propose the hypotheses test as it is shown in Equation (38), considering the statistics $\zeta = \{\gamma, \eta\}$ in a way that the acceptance domain $\mathcal{H}$ is defined for value ζ_0. In the following relation, $\mathcal{D}$ is a compact set where the values of the parameters are obtained from it.

$$\mathcal{H}(a) = \begin{cases} 1 & \text{if } \zeta \leq \zeta_0 \\ \dfrac{\displaystyle\int_{\zeta \leq \zeta_0} f(a; \gamma, \eta) d\zeta}{\displaystyle\int_{\mathcal{D}} f(a; \gamma, \eta) d\zeta} & \text{otherwise} \end{cases} \tag{38}$$

VI. A Reference Architecture for Supporting Clustering Validation via Non-Negative Matrix Factorization Trace Sequences over Probabilistic Spaces

In this Section, we introduce the proposed reference architecture designed to support clustering validation. Indeed, the architectural point-of-view of the proposed technique is really important, due to the fact that *performance* is a critical aspect of every data-intensive/big-data processing system (e.g., [7], [41]).

This architecture harnesses and leverages the power of non-negative matrix factorization trace sequences, demonstrating their efficiency within the context of complex probabilistic spaces. Our detailed exploration will shed light on the intricacies of this innovative approach and its potential to advance the field of clustering validation. Figure 4 serves as an illustrative representation of this comprehensive reference architecture.

As presented in Figure 4, this architecture encompasses various layers that cooperate to enable clustering validation via non-negative matrix factorization trace sequences. Specifically, these layers and components are briefly described next and, later, deeply discussed in the following Sections:

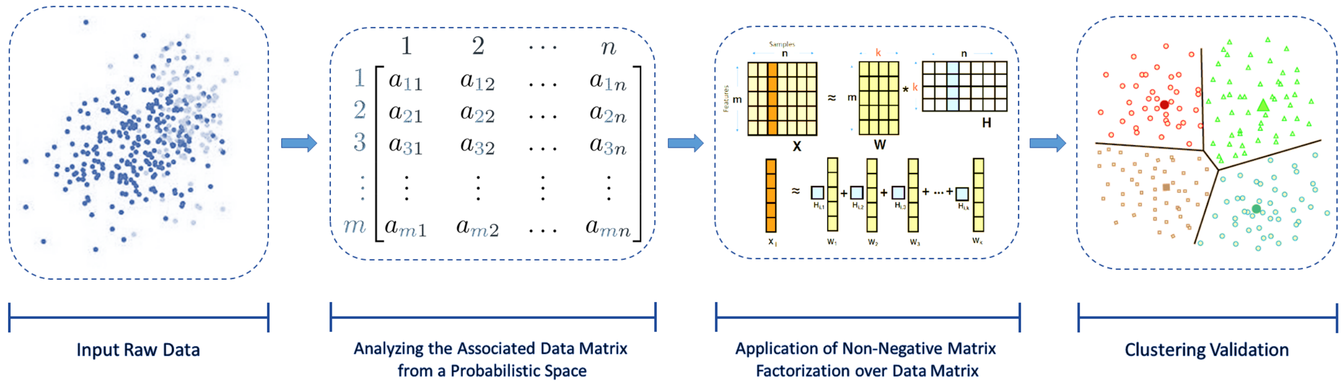

Fig. 4. Supporting Clustering Validation via Non-Negative-Matrix-Factorization Trace Sequences: A Reference Architecture

- *Input Raw Data*: this layer refers to the initial stage of the framework. Its primary purpose is to collect and receive data from various sources;
- *Analyzing the Associated Data Matrix From a Probabilistic Space*: this layer forms a vital part of the whole architecture, since it is responsible for transforming the raw data into a data matrix and also involves examining the derived data matrix from a probabilistic model in order to assess the relationships and patterns within the data;
- *Application of Non-Negative Matrix Factorization over Data Matrix*: this is the core layer of the proposed reference architecture - here, the data matrix is decomposed into non-negative factors using the non-negative matrix factorization technique;
- *Clustering Validation*: it is the layer where the quality and effectiveness of a clustering algorithm results are assessed. This process involves various metrics and techniques to measure the consistency and separation of clusters within the data.

In the next, we will focus on these layers and components with a deeper elaboration and presentation of more extensive information.

A. Input Raw Data

This layer represents the first layer of our architecture, which consists of the input raw data. This raw data serves as the foundational building component upon which our entire system is constructed. Specifically, it is responsible for the role of information sources in subsequent layers of processing.

B. Analyzing the Associated Data Matrix from a Probabilistic Space

It is the layer responsible for the fundamental procedure of analyzing the data after transformation into the associated data matrix. This data matrix typically represents the raw data in a tabular format, where rows represent individual observations (samples) and columns represent variables or features. Each cell in the matrix contains a measure corresponding to a specific observation or variable. Basically, this layer embodies the transformation of raw data into actionable information.

After getting the associated data matrix, the next phase consists of analyzing the data from a probabilistic space, which refers to the process of applying and leveraging probabilistic methods and techniques (e.g., *probability distributions*, *Bayesian statistics*, *probabilistic modeling*, etc.) to gain a deeper understanding of the data represented in the data matrix.

C. Application of Non-Negative Matrix Factorization over Data Matrix

The *application of non-negative matrix factorization over data matrix* layer serves as the core essence of this architecture, as it is involved in the process of evaluating and validating the quality and effectiveness of clustering algorithms applied to the data.

At its core, this layer involves the utilization of non-negative matrix factorization, a powerful mathematical technique used for dimensionality reduction and feature extraction. The primary objective of applying NMF in this layer is to transform the original data matrix, often representing high-dimensional data, into two lower-dimensional matrices, typically referred to as the basis and coefficient matrices, which have the property of non-negativity. This approach is particularly useful for feature extraction and dimensionality reduction while ensuring interpretability for various data analysis tasks (e.g., in our case, *clustering*).

D. Clustering Validation

It represents the layer that assumes elevated significance within the architecture, as it not only evaluates the quality of clustering results but also provides valuable insights into the credibility and probabilities associated with selecting the appropriate number of clusters via a Bayesian interpretation.

Furthermore, in the Bayesian interpretation of clustering validation, we treat the choice of the number of clusters (k) as a *model selection problem*. Instead of relying solely on traditional metrics, we employ *Bayesian probabilistic models* to assess the suitability of various k values. Another powerful aspect of Bayesian clustering validation is model averaging. Instead of selecting a single k value, model averaging considers a weighted combination of clustering solutions across different k values based on their posterior probabilities.

VII. Conclusions and Future Work

This paper discusses how NMF is applied in a space where the dimensions are defined within the domain of positive real numbers $\mathbb{S}_+$. In this context, NMF facilitates the creation of a sequence of traces, where each subsequent trace decreases in index. By normalizing this sequence, the resulting distribution approaches a Gamma PDF. This discovery has implications for solving the clustering validation problem and provides a validation criterion. A reference architecture implementing the technique has also been provided. Future work is mainly oriented towards extending our proposed theoretical framework as to deal with innovative aspects dictated by emerging *big data trends* (e.g., [4], [6], [9], [23]–[26]).

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